

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#04)

Use of Venus Simulator for Register Level Execution of RISC-V Arithmetic and Memory Instructions

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MARKS AWARDED: _____ / 10

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Task 1 :

Description:

In this 1st task of lab, we were asked to write basic code of assembly and perform arithmetic operations which include addition, subtraction, multiplication and power. Then write the formats and no. of bits each Part of code takes.

Code:

```
addi s0,x0,5 # value of A = 5
addi s1,x0,4 # value of B = 4
add s2,s0,s1 # value of C = 9
sub s3,s0,s1 # value of D = 1
mul s4,s0,s1 # value of E = 20
mul s5,s0,s0 # value of F = 25
```

Instruction	Format	Immediate	funct7	rs2	rs1	funct3	rd	Opcode
addi s0,x0,5	I	5	-	-	0	0	8	19
addi s1,x0,4	I	4	-	-	0	0	9	19
add s2,s0,s1	R	-	0	9	8	0	18	51
sub s3,s0,s1	R	-	32	9	8	0	19	51
mul s4,s0,s1	R	-	1	9	8	0	20	51
mul s5,s0,s0	R	-	1	8	8	0	21	51

PC	Machine Code	Basic Code	Original Code
0x0	0x00500413	addi x8 x0 5	addi s0,x0,5 # value of A = 5
0x4	0x00400493	addi x9 x0 4	addi s1,x0,4 # value of B = 4
0x8	0x00940933	add x18 x8 x9	add s2,s0,s1 # value of C = 9
0xc	0x409409B3	sub x19 x8 x9	sub s3,s0,s1 # value of D = 1
0x10	0x02940A33	mul x20 x8 x9	mul s4,s0,s1 # value of E = 20
0x14	0x02840AB3	mul x21 x8 x8	mul s5,s0,s0 # value of F = 25

s0 (x8)	5
s1 (x9)	4

s2 (x18)	9
s3 (x19)	1

s4 (x20)	20
s5 (x21)	25

Task 2:

Description:

In this 2nd task of lab, we were given a equation to solve. So in this task we use the previous code of task 01 and simply extend it to get the result. The main thing in this task was we cant write it like c language we have to perform step by step but this becomes more easy.

Code:

```
1 addi s0,x0,5 # value of A = 5
2 addi s1,x0,4 # value of B = 4
3 add s2,s0,s1 # value of C = 9
4 sub s3,s0,s1 # value of D = 1
5 mul s4,s0,s1 # value of E = 20
6 mul s5,s0,s0 # value of F = 25
7
8 sub s6,s0,s2
9 add s7,s3,s4
10 mul s8,s7,s5
11 add s9,s6,s8 # Final value of equation
```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x00500413	addi x8 x0 5	addi s0,x0,5 # value of A = 5
0x4	0x00400493	addi x9 x0 4	addi s1,x0,4 # value of B = 4
0x8	0x00940933	add x18 x8 x9	add s2,s0,s1 # value of C = 9
0xc	0x409409B3	sub x19 x8 x9	sub s3,s0,s1 # value of D = 1
0x10	0x02940A33	mul x20 x8 x9	mul s4,s0,s1 # value of E = 20
0x14	0x02840AB3	mul x21 x8 x8	mul s5,s0,s0 # value of F = 25
0x18	0x41240B33	sub x22 x8 x18	sub s6,s0,s2
0x1c	0x01498BB3	add x23 x19 x20	add s7,s3,s4
0x20	0x035B8C33	mul x24 x23 x21	mul s8,s7,s5
0x24	0x018B0CB3	add x25 x22 x24	add s9,s6,s8 # Final value of equation

s2 (x18)	9
s3 (x19)	1
s4 (x20)	20
s5 (x21)	25

s6 (x22)	-4
s7 (x23)	21
s8 (x24)	525
s9 (x25)	521

s0 (x8)	5
s1 (x9)	4

Task 3:

Description:

In this task 3 of lab, we were given two numbers and asked to perform logical operations on it. We perform **and**, **or**, **xor** operation on it. In **and** if any entry is 0 it gives 0 value, in **or** if any any entry is one it gives one and in **xor** same entries will give 0 and opposite gives 1.

Code:

```
1 .text
2
3 main:
4     li s0, 0xFFF
5     li s1, 0xF0F
6
7     and s2, s0, s1
8     or  s3, s0, s1
9     xor s4, s0, s1
```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x00001437	lui x8 1	li s0, 0xFFF
0x4	0xFFFF40413	addi x8 x8 -1	li s0, 0xFFF
0x8	0x000014B7	lui x9 1	li s1, 0xF0F
0xc	0xF0F48493	addi x9 x9 -241	li s1, 0xF0F
0x10	0x00947933	and x18 x8 x9	and s2, s0, s1
0x14	0x009469B3	or x19 x8 x9	or s3, s0, s1
0x18	0x00944A33	xor x20 x8 x9	xor s4, s0, s1

sp (x2)	2147483632
gp (x3)	268435456

s2 (x18)	3855
s3 (x19)	4095
s4 (x20)	240

Task 4:

Description:

In this task 4 of lab, the code was given in manual but with errors. So we were asked to understand the concept of loops and then make the corrections in the code so that it runs perfectly and produce the desire result.

Code:

```
.data
my_array:
    .word 5,10,15,20
.text
.globl arraysum
arraysum:
    # a0 = int a[]
    # a1 = int size
    # t0 = ret
    # t1 = i
    la    a0, my_array
    li    a1, 4
    li    t0, 0 # ret = 0
    li    t1, 0 # i = 0
1: # For loop
    bge   t1, a1, 1f    # if i >= size, break
    slli   t2, t1, 2 # Multiply i by 4 (1 << 2 = 4)
    add    t2, a0, t2    # Update memory address
    lw     t2, 0(t2) # Dereference address to get integer
    add    t0, t0, t2    # Add integer value to ret
    addi   t1, t1, 1 # Increment the iterator
    j      1b    # Jump back to start of loop (1 backwards)
1:
    mv     a0, t0    # Move t0 (ret) into a0
    ret    # Return via return address register
```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x10000517	auipc x10 65536	la a0, my_array
0x4	0x00050513	addi x10 x10 0	la a0, my_array
0x8	0x00400593	addi x11 x0 4	li a1, 4
0xc	0x00000293	addi x5 x0 0	li t0, 0 # ret = 0
0x10	0x00000313	addi x6 x0 0	li t1, 0 # i = 0
0x14	0x00B35063	bge x6 x11 0	bge t1, a1, 1f # if i >= size, break
0x18	0x00231393	slli x7 x6 2	slli t2, t1, 2 # Multiply i by 4 (1 << 2 = 4)
0x1c	0x007503B3	add x7 x10 x7	add t2, a0, t2 # Update memory address
0x20	0x0003A383	lw x7 0(x7)	lw t2, 0(t2) # Dereference address to get integer
0x24	0x007282B3	add x5 x5 x7	add t0, t0, t2 # Add integer value to ret
0x28	0x00130313	addi x6 x6 1	addi t1, t1, 1 # Increment the iterator
0x2c	0xFE9FF06F	jalt x0 -24	j 1b # Jump back to start of loop (1 backwards)
0x30	0x00028513	addi x10 x5 0	mv a0, t0 # Move t0 (ret) into a0
0x34	0x00000000	addi x10 x10 0	ret # Return into caller's world

t0 (x5)	50
t1 (x6)	4
t2 (x7)	20

Task 5:

Description:
In the lab task 5, we were asked to implement the concept of storing in memory and then loading the same data from memory. And we are specifically asked to use lw command to load data and .data & .word commands to store the data. This task is performed by repeated the same concept for each value of table
Code:

```
.data
table:
    .word 2,4,6,8,10,12,14,16,18,20

.text
.globl table_display

    la s0,table
    lw s0,0(s0)
    la s1,table
    lw s1,4(s1)
    la s2,table
    lw s2,8(s2)
    la s3,table
    lw s3,12(s3)
    la s4,table
    lw s4,16(s4)
    la s5,table
    lw s5,20(s5)
    la s6,table
    lw s6,24(s6)
    la s7,table
    lw s7,28(s7)
    la s8,table
    lw s8,32(s8)
    la s9,table
    lw s9,36(s9)
```

Output:

PC	Machine Code	Basic Code	Original Code
0x0	0x10000417	auipc x8 65536	la s0,table
0x4	0x00040413	addi x8 x8 0	la s0,table
0x8	0x00042403	lw x8 0(x8)	lw s0,0(s0)
0xc	0x10000497	auipc x9 65536	la s1,table
0x10	0xFF448493	addi x9 x9 -12	la s1,table
0x14	0x0044A483	lw x9 4(x9)	lw s1,4(s1)
0x18	0x10000917	auipc x18 65536	la s2,table
0x1c	0xFE890913	addi x18 x18 -24	la s2,table
0x20	0x00892903	lw x18 8(x18)	lw s2,8(s2)
0x24	0x10000997	auipc x19 65536	la s3,table
0x28	0xFDC98993	addi x19 x19 -36	la s3,table
0x2c	0x00C9A983	lw x19 12(x19)	lw s3,12(s3)
0x30	0x10000A17	auipc x20 65536	la s4,table
0x34	0xFD0A0A13	addi x20 x20 -48	la s4,table
0x38	0x010A2A03	lw x20 16(x20)	lw s4,16(s4)
0x3c	0x10000A97	auipc x21 65536	la s5,table
0x40	0xFC4A8A93	addi x21 x21 -60	la s5,table
0x44	0x014AAA83	lw x21 20(x21)	lw s5,20(s5)
0x48	0x10000B17	auipc x22 65536	la s6,table
0x4c	0xFB8B0B13	addi x22 x22 -72	la s6,table
0x50	0x018B2B03	lw x22 24(x22)	lw s6,24(s6)
0x54	0x10000B97	auipc x23 65536	la s7,table
0x58	0xFACB8B93	addi x23 x23 -84	la s7,table
0x5c	0x01CBAB83	lw x23 28(x23)	lw s7,28(s7)
0x60	0x10000C17	auipc x24 65536	la s8,table
0x64	0xFA0C0C13	addi x24 x24 -96	la s8,table
0x68	0x020C2C03	lw x24 32(x24)	lw s8,32(s8)
0x6c	0x10000C97	auipc x25 65536	la s9,table
0x70	0xF94C8C93	addi x25 x25 -108	la s9,table
0x74	0x024CAC83	lw x25 36(x25)	lw s9,36(s9)

s0 (x8)	2
s1 (x9)	4
s2 (x18)	6
s3 (x19)	8
s4 (x20)	10
s5 (x21)	12
s6 (x22)	14
s7 (x23)	16
s8 (x24)	18
s9 (x25)	20